

Red Hat Lightspeed 1-latest

Assessing and monitoring security vulnerabilities on RHEL systems with FedRAMP

Understanding your environmental exposure to potential security threats

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Understanding your environmental exposure to potential security threats

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Abstract

Use the vulnerability service with FedRAMP® to assess and monitor the status of security vulnerabilities on your Red Hat Enterprise Linux systems, understand the level of exposure of your infrastructure, and plan a course of action.

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CHAPTER 1. OVERVIEW OF THE VULNERABILITY SERVICE

The vulnerability service enables quick assessment and comprehensive monitoring of the exposure of your RHEL infrastructure to Common Vulnerabilities and Exposures (CVEs). Use this information to better understand your most critical issues and systems and effectively manage remediations.

1.1. PREREQUISITES

- Each system has the insights-client installed and registered to the Red Hat Lightspeed application. Follow the *Getting Started with Red Hat Lightspeed* guide to install the client and register your system(s).
- The vulnerability service is fully supported for RHEL systems managed by Red Hat Subscription Management (RHSM) and Satellite 6 and later. Using any other means to obtain package updates can lead to misleading results. Only use Satellite 6 with RHSM or RHSM registered with subscription.redhat.com (Customer Portal).
- Vulnerability service remediations are not fully supported and might not work properly on Satellite 5 and Spacewalk-hosted RHEL systems.
- Some features require special privileges provided by your organization administrator. The ability to view Red Hat Security Advisories (RHSAs) associated with certain CVEs and systems requires permissions granted through user access. The ability to view and patch those vulnerabilities in the Red Hat Lightspeed patch service also requires permissions granted through user access.

1.2. HOW THE VULNERABILITY SERVICE WORKS

The vulnerability service uses the insights-client to gather information about your RHEL systems. The client gathers information about the systems and uploads it to the vulnerability service.

After the client gathers information, the vulnerability service assesses the data against the Red Hat CVE database and security bulletins. This allows the vulnerability service to determine if there are any outstanding CVEs that could affect the systems, and provides the results of those comparisons.

Once the vulnerability service analyzes the data, you can view and sort the displayed results. You can also assess the risks and priorities of the vulnerabilities, report their status, and create and deploy Ansible Playbooks to remediate them. The goal of the vulnerability service is to enable a repeatable process that protects against security weaknesses in your RHEL infrastructure.



IMPORTANT

The vulnerability service shows all CVEs available for the major version of your RHEL system, even if the system has its release version locked to a specific minor release. To find if the CVE was applied to the minor version of your RHEL system, in the **Security → Vulnerability → Systems**, locate your system. The **Remediation type** must be **Manual**.

1.3. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

1.3.1. User Access overview

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

All users on your account have access to most of the data in Red Hat Lightspeed.

Additional resources

- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP

1.3.2. Predefined groups in User Access

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

1.3.3. Predefined roles assigned to groups

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

TIP

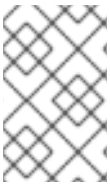
Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Predefined User Access roles

1.3.4. Check your permissions

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.

2. If you try to access Red Hat Lightspeed features and see a message that you do not have

permission to perform this action, contact your Organization Administrator or a user with the **User Access administrator** role to request the permissions required to access those features and complete the actions you want to perform.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

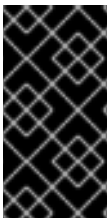
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

1.3.5. Configure user permissions

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features
- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

1.3.6. User Access roles for permissions to vulnerability service features

Understand the predefined roles that control access to features of the vulnerability service in Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

The following roles enable enhanced access to vulnerability features:

Table 1.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...	Included in the Default access group
Vulnerability administrator	• Do everything that a **Vulnerability viewer** can do. • Do any available operations against any vulnerability-service resource.	
Vulnerability viewer	• Read any vulnerability-service resource.	X

CHAPTER 2. COMMON VULNERABILITIES AND EXPOSURES (CVES)

Common Vulnerabilities and Exposures (CVEs) are security vulnerabilities identified in publicly released software packages. The vulnerability service includes only those CVEs for which Red Hat issues security advisories (RHSAs).

2.1. RED HAT SECURITY ADVISORIES (RHSAS)

Red Hat Security Advisory (RHSA) errata document security vulnerabilities in Red Hat products for which there are remediations or mitigations available. The Red Hat Lightspeed vulnerability service displays the advisory identifier tied to each system exposed to a CVE.

View this information by selecting a CVE and selecting the **Filter by affected systems** link in the security rule card. If an advisory exists for the system, the RHSA ID is visible as a link next to the system in the **Exposed systems** list, **Advisory** column. When there are no such advisories, the Advisory column is not visible, or shows **Not available**.

When an advisory exists for a system, users can view more information about the RHSA, including a list of affected systems. In the patch service, users can select systems to create an Ansible Playbook to apply the remediation.

2.2. SECURITY RULES

Security rules are CVEs given additional visibility due to the elevated risk and exposure associated with them.

These are security flaws that can receive significant media coverage. The Red Hat Product Security team provides Product Security Incident Response Plan workflow, to help find your RHEL environment exposure. These security rules allow you to take the appropriate action to protect your organization.

Security rules provide deep threat intelligence, beyond analyzing the version of RHEL running on a system. Security rules are manually curated to find whether you are susceptible to a security threat by analyzing system metadata collected by the insights-client. If the vulnerability service identifies a system as exposed to a security rule, there is the potential for elevated security risk and you must address this issues with urgency.



IMPORTANT

Addressing security rules on exposed systems should be your highest priority.

Not all systems exposed to a CVE are also exposed to a security rule associated with that CVE. Even though you might be running a vulnerable version of software, other environmental conditions might mitigate the threat. Examples include a specific port is closed or running SELinux.

Additional resources

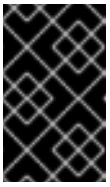
- Product Security Incident Response Plan

2.2.1. Identify security rules in the Red Hat Lightspeed dashboard

You can view your infrastructure exposure to security rules to identify a potential threat to your systems.

Procedure

1. Navigate to the Red Hat Lightspeed dashboard.
2. View the **Latest critical notifications** on your systems panel.
   These are security rules with an elevated severity rating of "Important" or "Critical." These are potentially your most critical issues and you must prioritize them for remediation.
   a. To the right of each notification, click the **Expand** button to see associated CVEs and the number of systems exposed in your infrastructure.



IMPORTANT

You might see security rules in your critical notifications but have zero systems exposed. In this case, even though the CVE is present in your infrastructure, the security rule conditions may not exist.

   b. Below the name of the security rule, and under Associated CVEs, click the CVE ID link.
   c. View which of your systems the security rule CVEs impact, and, optionally, select exposed systems to create playbooks.
3. Next, view the information in the **vulnerability** card.
   a. Note the number of "CVEs with **security rules** impacting systems."
   This number includes security rules of any severity impacting at least one system.
      i. Click **View CVEs**.
      Consider lesser-severity security rules your second highest priority for remediation, following high-severity security rules.

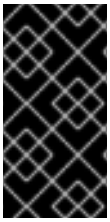
Additional resources

- Product Security Incident Response Plan
- Generating vulnerability service reports

2.3. KNOWN EXPLOITS

Red Hat analyzes Cybersecurity and Infrastructure Security Agency (CISA) data, to decide whether code exists publicly to exploit a CVE, or a CVE has already been exploited publicly. The vulnerability service applies the "Known exploits" label to CVEs that meet that criteria.

This enhanced threat assessment helps users identify and address those CVEs that pose the most critical risks first. Red Hat recommends that users review CVEs with the **Known exploit** label with high priority and work towards remediating those issues.



IMPORTANT

The vulnerability service makes you aware that the known-exploit CVE exists on systems in your infrastructure. The **Known exploits** label does not mean that the vulnerability was exploited on your RHEL systems; the vulnerability service does not make that determination.

2.4. COMMON VULNERABILITIES AND EXPOSURES PROVIDE DEEP THREAT INTELLIGENCE WITH TRIAGE FEATURE

The vulnerability service provides you with data about individual Common Vulnerabilities and Exposures (CVEs) and their effect on your systems registered to Red Hat Lightspeed.

CVEs are categorized as **vulnerable** or **affected but not vulnerable**. This level of threat intelligence is available for CVEs that have the **Security Rule** label or for those that have gone through Red Hat Product Security analysis.

This increased threat intelligence enables you to triage issues and address the most urgent ones first. When managing a large fleet of servers, this translates into expedited protection and significant efficiencies.

An **affected but not vulnerable** CVE status indicates that you are running software that has a vulnerability in it but is not currently exploitable. This system needs remediation but does not require immediate attention.

A **vulnerable** CVE status indicates flawed code with an open path to exploitation. An open path could be a port or an OS version that permits one of the following:

- confidential information to be leaked,
- the integrity of the system to be compromised,
- availability of the system to be hindered.

Examine an example of a **vulnerable** server versus an **affected but not vulnerable** server:

Suppose that **Server A** is running vulnerable software that allows root access to the system. **Server A** would be considered vulnerable and require immediate patching.

In contrast, suppose that **Server B's** current configuration prevents the vulnerability from manifesting, even when present in the affected code. **Server B** would be considered **affected but not vulnerable**. This would mean that **Server B** could be relegated to the to-do list, so that the more immediate threat, **Server A** could be remediated.



IMPORTANT

You should patch **Server B** once **Server A** has been addressed since it is running potentially vulnerable code. Version updates and other events could render it vulnerable in the future.

2.4.1. Identify known-exploit CVEs in the Red Hat Lightspeed dashboard

You can identify known-exploit CVEs in the Red Hat Lightspeed dashboard vulnerability card.

Procedure

1. Navigate to the Red Hat Lightspeed dashboard.
2. On the **Vulnerability** card, note the **CVEs with Known exploits impacting 1 or more systems and the number displayed**.
3. Click **View Known exploits**.

The filtered list of Known-exploit CVEs in the CVEs list displays.

Additional resources

- CISA Known Exploited Vulnerabilities Catalog

2.5. ADDITIONAL RESOURCES

- For more information, see the list of CVEs

CHAPTER 3. REFINING VULNERABILITY-SERVICE RESULTS

The vulnerability service enables many ways to refine the views of your data, helping you and others focus on your most critical systems, workloads, or issues.

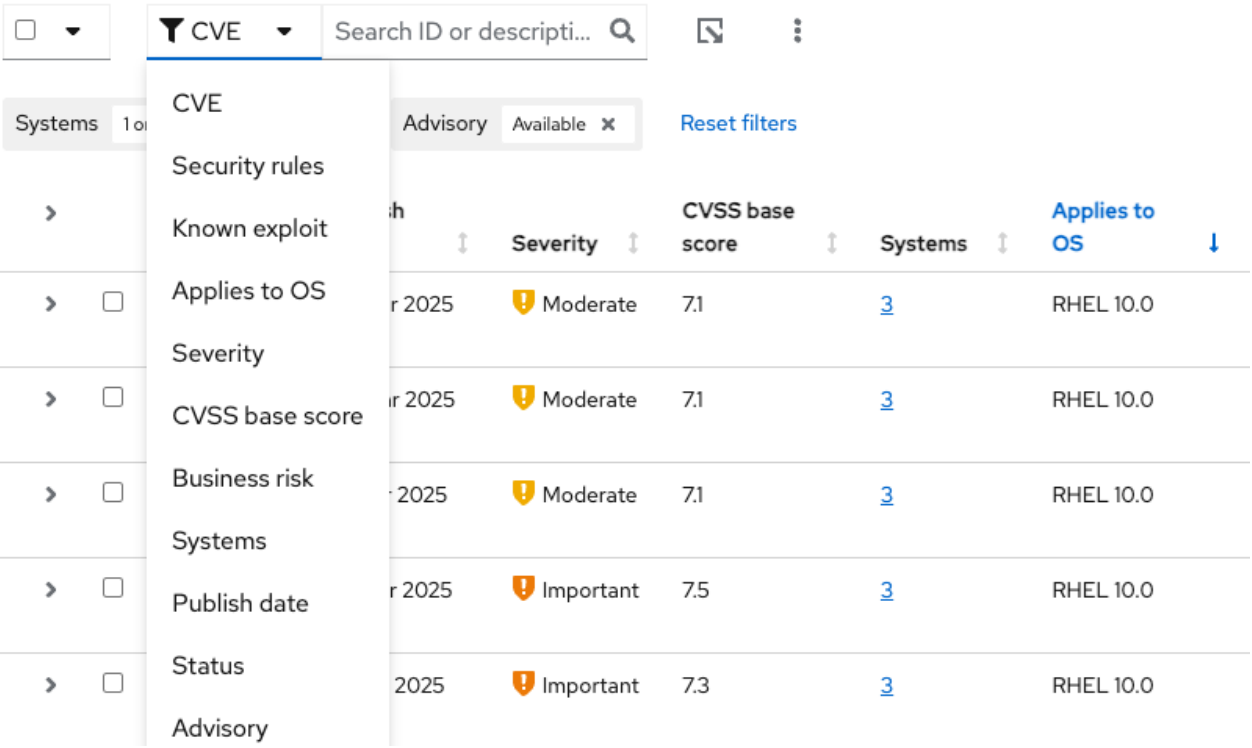
3.1. CVE-LIST AND SYSTEMS-LIST FILTERS

Filtering narrows the visible list of CVEs and associated systems, helping you to focus on specific issues.

You can apply filters to the CVEs list to focus on CVEs by criticality or business risk, for example. After selecting an individual CVE, apply filters to the resulting list of affected systems to focus on those of a specific RHEL major or minor version, for example.

To activate filters, select a primary filter from the drop-down list of filters on the left. Then select a secondary subfilter from the filter options drop-down list on the right. Selected filters are visible below the Filters menu. To deactivate filters, click the X next to each one.

CVEs list filters

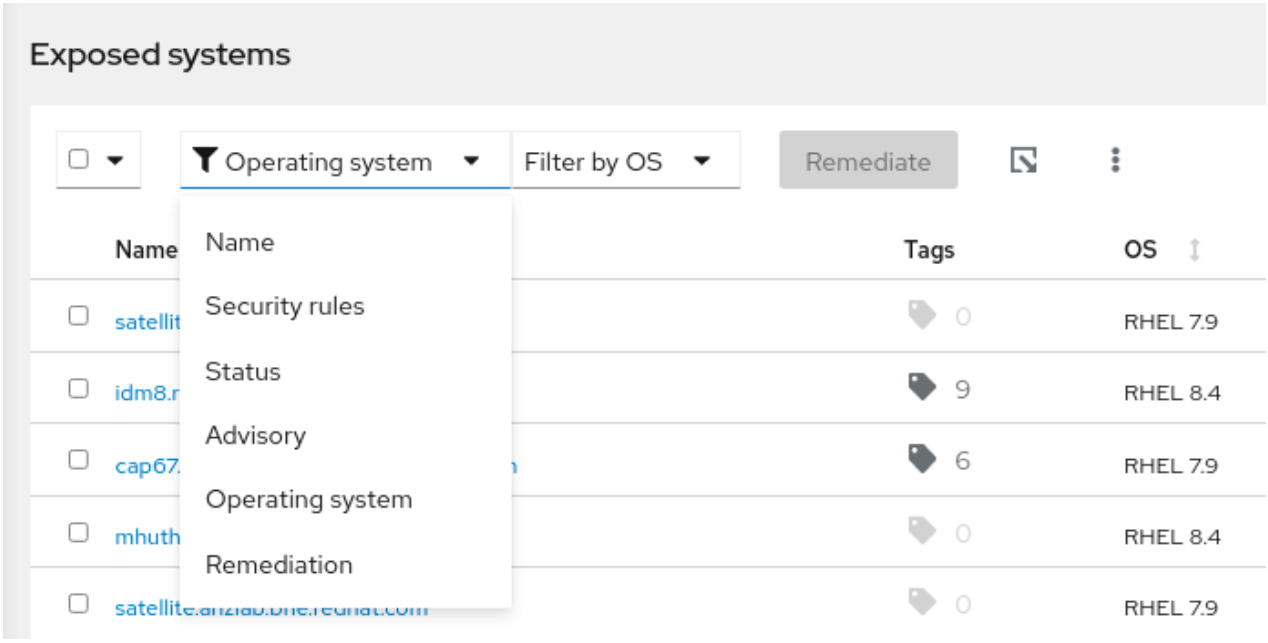


The following primary filters are accessible from the CVEs page. Select the primary filter, then define a parameter in the subfilter:

- **CVE.** Search ID or description.
- **Security rules.** Show only CVEs with the **Security rule** label.
- **Known exploit.*** Show only CVEs with the **Known exploit** label.
- **Severity.** Select one or more values: Critical, Important, Moderate, Low, or Unknown.
- **Applies to OS.** Select up to five minor versions of Red Hat Enterprise Linux to view vulnerability exposure for those versions to a specific CVE.
- **CVSS base score.** Select one or more ranges: All, 0.0-3.9, 4.0-7.9, 8.0-10.0, N/A (not applicable)

- **Business risk.** Select one or more values: High, Medium, Low, Not defined.
- **Systems exposed.** Select to only show CVEs with systems currently affected, or with no systems affected.
- **Publish date.** Select from All, Last 7 days, Last 30 days, Last 90 days, Last year, or More than 1 year ago.
- **Status.** Select one or more values: Not reviewed, In review, On-hold, Scheduled for patch, Resolved, No action - risk accepted, Resolved via mitigation.

Systems list filters

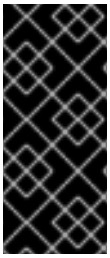


The following primary filters are accessible from the top of the list of systems on the CVE details page:

- **Name.** Find a specific CVE by entering the CVE ID.
- **Security rules.** If the CVE has a security rule associated with it, filter by other systems vulnerable to the same security rule, or show systems not affected by the security rule.
- **Status.** Show systems in specific status or workflow categories.
- **Advisory.** Show systems to which a Red Hat advisory applies for this CVE.
- **Operating system.** Show systems running specific RHEL (minor) versions.
- **Remediation.** Show systems included in an Ansible Playbook, a manual remediation, or that are not included in a current remediation plan.

3.1.1. Filter security-rule CVEs

High-severity security rules pose the greatest threat to your infrastructure. Prioritize identifying and remediating these CVEs. You can filter the CVEs list to show only high-severity security-rule CVEs to identify affected systems.



IMPORTANT

Not all systems exposed to a CVE are also exposed to a security rule associated with that CVE. Even though you might be running a vulnerable version of software, other environmental conditions might mitigate the threat. For example, a specific port is closed or an enabled SELinux policy might prevent the CVE from being exploited on your system.

Procedure

1. Navigate to Security > Vulnerability > CVEs in Red Hat Lightspeed.
2. Click the filters dropdown list in the toolbar.
   a. Apply the **Security rules** primary filter.
   b. Apply the **Has security rule** subfilter.
3. Scroll down to view security-rule CVEs.
   CVEs with security rules display the security-rule label located immediately below the CVE ID.

Additional resources

- Product Security Incident Response Plan
- Understanding severity ratings

3.1.2. Remediate vulnerabilities with security rules on RHEL Systems

CVEs with security rules are CVEs prioritized by Red Hat because they focus on issues that have elevated risk to your systems. Remediating these issues helps support a security posture that prioritizes the most important issues for your organization.

Using the vulnerability service and the remediations service, you can prioritize and remediate some of the most important threats to your systems by:

- Focusing on CVEs that have security rules.
- Remediating CVEs.

Additional resources

- Security rules
- Filtering lists of systems exposed to security rules
- Red Hat Lightspeed remediations guide with FedRAMP

3.1.3. Filter known-exploit CVEs

Red Hat recognizes and labels CVEs that have widely known exploits with the **Known exploit** label. Known exploits either have a documented proof of concept, or have already been used to attack vulnerable systems.

CVEs with the **Known exploit** label pose a more immediate threat to your infrastructure. As a best practice, prioritize identifying and remediating these CVEs.



IMPORTANT

Red Hat does not decide whether any of your registered systems have been exploited. Red Hat identifies CVEs that might pose an serious risk.

You can filter known-exploit CVEs in the CVEs list.

Procedure

1. Navigate to Security > Vulnerability > CVEs in Red Hat Lightspeed.
2. Click the filters drop-down list in the toolbar.
   a. Apply the **Known exploit** filter.
   b. Apply the **Has a known exploit** subfilter.
3. Scroll down to view a list of CVEs with known exploits.

Additional resources

- CISA Known Exploited Vulnerabilities Catalog
- Understanding severity ratings

3.1.4. Filter CVEs without associated advisories

Some CVEs do not have associated advisories, also called *errata*. This might happen for any of the following reasons:

- No fix is available for the CVE.
- Product Security analysis determines that the CVE affects your environment, but has no errata available for your environment. The same CVE can have errata in other environments.
- Your system is no longer under support.



IMPORTANT

CVE information is currently available for RHEL 6 and newer.

You can identify the CVEs without advisories to take measures to protect your organization from exposures associated with those vulnerabilities. This allows you to take the necessary steps to address the issues.

If your version of RHEL does not have a fix available and has a “Will not fix” label, consider the following criteria:

- The impact of the vulnerability (severity)
- The life cycle phase of your version of RHEL

If you decide that a fix is necessary for a CVE without an associated advisory, the following options are available:

- Accept the risk
- Upgrade to a supported product version that includes a fix for this vulnerability, if available (recommended)
- Apply a mitigation (if one exists)

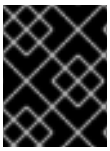
Additional resources

- Common Vulnerabilities and Exposures
- Understanding severity ratings
- Life cycle and update policies
- Customer support

3.1.4.1. Enable CVEs without advisories

You can enable CVEs without advisories to access systems affected by CVEs without advisories in Red Hat Lightspeed.

This feature is enabled by default, but CVEs without advisories are hidden by default in the main view. This means that you must use filters to display and view CVEs without advisories.



IMPORTANT

Red Hat's policy requires Red Hat Lightspeed to display all high priority, critical, and important CVEs, regardless of whether those CVEs have associated advisories.

Prerequisites

- Vulnerability administrator access to your environment in Red Hat Lightspeed

Procedure

1. From the Red Hat Lightspeed dashboard, navigate to **Security → Vulnerability → CVEs**.
2. Click the **More options** icon (⋮) and select **Show CVEs without Advisories**. The list of advisories includes CVEs without advisories.

Additional resources

- Understanding severity ratings
- Life cycle and update policies

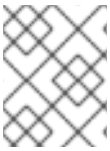
Verification

- The CVE list now includes CVEs without advisories when you apply the **Filter by Advisory** > **Not Available** filter.

3.1.4.2. Disable CVEs without advisories

You can disable the **CVEs without advisories** feature by deselecting the **Show CVEs without Advisories** option.

The **CVEs without advisories** option is enabled by default, but the default view hides CVEs without advisories.



NOTE

Red Hat policy requires Red Hat Lightspeed to display all high priority, critical, and important CVEs, regardless of whether those CVEs have associated advisories.

Prerequisites

- Vulnerability administrator access to your environment in Red Hat Lightspeed
- The list of advisories includes CVEs without advisories

Procedure

1. From the Red Hat Lightspeed dashboard, navigate to **Security → Vulnerability → CVEs**.
2. Click the **More options** icon (⋮) and select **Hide CVEs without Advisories**.

Additional resources

- Understanding severity ratings
- Life cycle and update policies

Verification

- CVEs without advisories no longer appear in the CVE list, even when using the **Filter by Advisory → Not Available** filter.

3.1.4.3. View CVEs without advisories

The **Show CVEs without Advisories** option enables or disables CVEs without advisories. To view CVEs without advisories, you must enable the **Show CVEs without Advisories** option.

Prerequisites

- The Organization Administrator has enabled the CVEs without advisories option.

Procedure

1. From the Red Hat Lightspeed dashboard, navigate to **Security → Vulnerability → CVEs**.
2. From the filter drop-down, select **Advisory**.
3. From the **Filter by Advisory** drop-down, select **Not Available**.
   The list of advisories shows all CVEs without advisories.

Additional resources

- Understanding severity ratings

3.1.4.4. Identify systems affected by a CVE without advisories

The CVE details page displays the list of all systems affected by the selected CVE. You can filter the list of systems to display the systems affected by the CVE where an advisory is not present.

Prerequisites

- The Organization Administrator has enabled the CVEs without advisories option.

Procedure

1. Identify a CVE without advisories for which you want to see systems that it affects. For more information about identifying CVEs without advisories, refer to Identifying systems with CVEs without Advisories.
2. Select the identified CVE to navigate to the CVE details page.
   The CVE details page for that CVE displays. The page lists all the systems affected by that CVE.
   a. If you apply the **Filter by Advisory** filter and **Not Available** option when you select a CVE, these filters persist to the CVE details page.
   b. Otherwise, when you navigate to the CVE details page, select **Advisory** from the filters at the top of the page.
   c. Select **Filter by Advisory**, and click the **Not Available** checkbox.
      The list of systems updates to show only systems affected by that CVE without advisories. The **Advisory** column shows **Not Available** for each system in the list.
3. Optional: To view details for the system, select the name of the system you want to view.
   The details page for the system displays.

Additional resources

- Understanding severity ratings

3.1.4.5. View CVEs without advisories in system details

The system details page displays the list of all CVEs that affect the selected system. You can filter the list of CVEs to display the CVEs without advisories.

Prerequisites

- The Organization Administrator has enabled the CVEs without advisories option.

Procedure

1. From the Red Hat Lightspeed dashboard, navigate to **Security → Vulnerability → Systems**.
   The Vulnerability systems page displays.
2. Select a system ID from the list.
   The system details page for that system opens. The page lists all the CVEs that affect the selected system.
3. Select **Advisory** from the filters at the top of the page.

4. Select **Filter by Advisory**, and select the **Not Available** checkbox.
   The list of CVEs updates to show only CVEs without advisories. The Advisory column shows **Not Available** for each CVE in the list.

5. Optional: To view details for the CVE, select the CVE ID for the CVE you want to view.
   The details page for the CVE displays.

Additional resources

- Understanding severity ratings

3.2. FILTER CVES BY RHEL VERSION

You can view and manage systems running a specific version of RHEL that are exposed to the selected CVE in the **Applies to OS** column on the CVEs page.

It lists the major and minor versions of RHEL that each CVE affects. By viewing this information, you can understand the degree of vulnerability exposure for those systems, and make informed decisions about possible upgrade paths.



NOTE

The vulnerability exposure feature supports RHEL major and minor versions from RHEL 6.9 to RHEL 10.1 and later.

Prerequisites

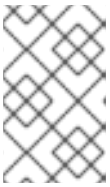
- You are logged in as an Organization Administrator, or you have Vulnerability Administrator access to your environment in Red Hat Lightspeed.

Procedure

1. From the Red Hat Lightspeed dashboard, navigate to **Security → Vulnerability → CVEs**.
   The CVEs page displays a list of CVEs that could affect systems in your environment.

2. Select the **Applies to OS** filter.

3. In the **Filters by Applies to OS** subfilter, click the drop-down menu and select the versions of RHEL you want to view.
   The list of CVEs displays the RHEL versions you selected.



NOTE

For a quick view of all RHEL versions affected by a specific CVE, navigate to the row that shows the CVE and hover your cursor over the ellipsis (...) in the **Applies to OS** column. A tooltip displays with a list of all affected RHEL versions.

Additional resources

- Generating vulnerability service reports

3.3. FILTER LISTS OF SYSTEMS EXPOSED TO SECURITY RULES

After filtering the list of CVEs to view only your most critical potential threats, select an individual CVE to view the list of exposed systems. You can apply a filter to the list.

Procedure

1. After selecting a security-rule CVE, scroll down to the **Exposed systems** list.
   Not every system in the list has the security rule conditions present for the CVE to be a security rule. Apply the following filter to see only the systems with security rule conditions present.
2. Select the **Security rules** filter from the primary filter drop-down list.
3. Check the **Has security rule** box in the secondary subfilter drop-down list.
4. View the systems with exposure to that CVE that also have the conditions present for the security rules.

Additional resources

- Product Security Incident Response Plan

3.4. RED HAT LIGHTSPEED GROUP FILTERS

You can view only those systems that belong to a specific group by filtering the vulnerability service results by groups of systems or workloads. These can be systems running SAP workloads or SAP ID, by Satellite host groups, or by custom tags added to the insights-client configuration file.

You can set group filtering globally in Red Hat Lightspeed using the **Filter results** box located at the top of the page throughout the Red Hat Lightspeed application. Group selection persists when changing from service to service and page to page. However, the functionality varies within the different Red Hat Lightspeed services.

Group filtering works in the vulnerability Dashboard and vulnerability service CVEs and Systems lists.

See *Tags and system groups* for information about group tags and configuring custom tags.

3.4.1. Filter Dashboard, CVEs, and Systems lists by group

You can filter vulnerability service CVE and Systems lists by group.

Procedure

1. Navigate to Red Hat Hybrid Cloud Console and log in.
2. Open the Red Hat Lightspeed application.
3. Click the down arrow on the **Filter results** box located at the top of any page in the Red Hat Lightspeed application.
4. Select a group by which to filter your systems.
   Search or scroll to view available tags. To browse the full list of available tags, scroll to the bottom of the list and click **View more**.

   Optional: Select additional filter options:

   a. Select SAP workloads.

b. Select systems by specific SAP IDs.

c. Select Satellite host collections.

d. Select systems identified by custom group tags.
For more information about creating custom tags, see **Custom system tagging**.

5. Navigate to the service and view only systems or CVEs that belong to your selected group or groups.

Additional resources

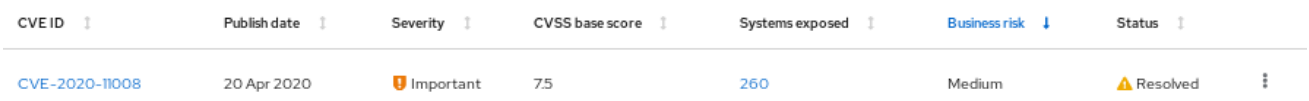
- System registration and client configuration

3.5. DEFINE A BUSINESS RISK FOR A CVE

You can define the business risk of a CVE by using the vulnerability service with the following options: High, Medium, Low, or Not Defined (default).

While the list of CVEs shows the severity of each CVE, assign a business risk to rank CVEs based on the impact they could have on your organization. This can give you more control in managing your risk efficiently in a large environment, and enable you to make better operational decisions.

By default, the business risk field for a specific CVE is set to **Not Defined**. After you set the business risk, it is visible in the Security > Vulnerability > CVEs list, in the CVE row.



Business risk is also visible on the details card for each CVE, which shows more information and lists affected systems.



3.5.1. Set a business risk level for a single CVE

You can set the business risk level for a single CVE by using the vulnerability service.



IMPORTANT

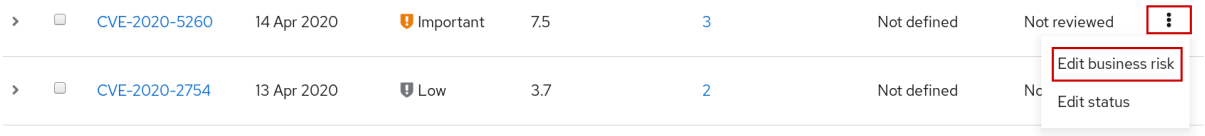
The business risk level for that CVE is the same on *all* systems impacted by it.

Prerequisites

- You are logged into the Hybrid Cloud Console.

Procedure

1. Navigate to the Security > Vulnerability > CVEs page..
2. Identify a CVE for which to set a business risk level.
3. Click the **more-actions** icon (three vertical dots) at the end of the CVE row, and then click **Edit business risk**.



4. Set a business risk level value to the appropriate level and, optionally, add a justification for your risk assessment.
5. Click **Save**.

Additional resources

- Understanding severity ratings
- Generating vulnerability service reports

Verification

- Locate the CVE in the list and verify the business risk level is set to the value you selected.

3.5.2. Set a business risk level for multiple CVEs

You can set the same business risk level on multiple CVEs that you select by using vulnerability service.

Prerequisites

- You are logged in to the Hybrid Cloud Console.

Procedure

1. Navigate to Security > Vulnerability > CVEs .
2. Select the CVEs for which you want to set a business risk level.
3. Click the **more-actions** icon (three vertical dots) to the right of the Filters drop-down menu in the toolbar and click **Edit business risk**
4. Set an appropriate business risk level value and, optionally, add a justification for your risk assessment.
5. Click **Save**.

Additional resources

- Understanding severity ratings
- Generating vulnerability service reports

Verification

- Verify the business risk level is set for all selected CVEs in the CVE list.

3.6. EXCLUDE SYSTEMS FROM VULNERABILITY SERVICE ANALYSIS

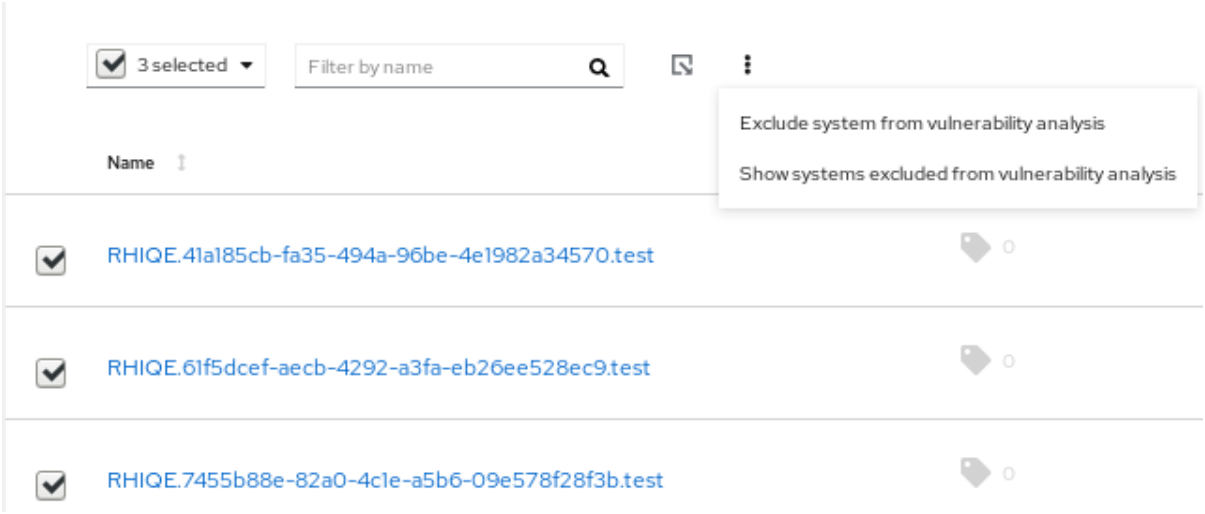
You can exclude specific systems from vulnerability analysis by using the vulnerability service. This can save you the time and attention required to review and re-review issues on systems that are not relevant to your organization's goals.

For example, if you have the following category of servers: QA, Dev, and Production, you might not care to review the vulnerabilities for your QA servers. And therefore want to exclude these systems from the analysis performed by the vulnerability service.

When you exclude systems from vulnerability analysis, the insights-client still runs as scheduled on the system, but the results for the system are not visible in the vulnerability service. The continued operations of the client ensure that other Red Hat Lightspeed services can still upload the data they need. It also means that you can still view results for those systems by using filtering.

Procedure

1. Navigate to the Security > Vulnerability > Systems tab and log in if necessary.
2. Check the box for each system you want to exclude from vulnerability analysis.
3. Click the **more-actions** icon in the toolbar, at the top of the list of systems, and select **Exclude systems from vulnerability analysis**.



4. Optional: You can exclude a single system by clicking the **more-actions** icon in the system row and selecting **Exclude system from vulnerability analysis**.



Additional resources

- System registration and client configuration

Verification

- Navigate to the Security > Vulnerability > Systems tab and verify the excluded systems no longer appear in the default systems list.
- To view excluded systems, click the **more-actions** icon and select **Show systems excluded from analysis**.

3.7. DISPLAY PREVIOUSLY EXCLUDED SYSTEMS

You can display a previously excluded system.

Procedure

1. Navigate to the Security > Vulnerability > Systems tab and log in if necessary.
2. Click the **more-actions** icon in the toolbar, at the top of the list of systems, and select **Show systems excluded from analysis**.

Verification

- Verify that your systems were excluded by observing the value of **Excluded** in the **Applicable CVEs** column.

Additional resources

- System registration and client configuration

3.8. RESUME VULNERABILITY ANALYSIS FOR A SYSTEM AFTER A PAUSE

You can resume vulnerability analysis for a system if a pause occurred.

Procedure

1. Navigate to the Security > Vulnerability > Systems tab and log in if necessary.
2. Click the **more-actions** icon in the toolbar, at the top of the list of systems, and then select **Show systems excluded from analysis**
3. In the list of results, select each system for which you want to resume vulnerability analysis.
4. Click the **more-actions** icon again and select **Resume analysis for system**

Additional resources

- System registration and client configuration

Verification

- Navigate to the Security > Vulnerability > Systems tab and verify the systems now appear in the default systems list.

3.9. CVE STATUS

You can set a status for CVEs to manage CVEs impacting your systems. The vulnerability service enables the following ways of setting a status for a CVE:

- Set a status for a CVE for *all* systems.
- Set a status for a *specific CVE + system pair* .

Status values are preset and include the following options:

- Not reviewed (default)
- In-review
- On-hold
- Scheduled for patch
- Resolved
- No action - risk accepted
- Resolved via mitigation

Setting a status for a CVE helps with better triaging through its life-cycle, from becoming aware of it to remediating it. Defining a status allows your organization to keep better tabs on where the most critical CVEs are in their life-cycle. It also focuses you on where your efforts must be to address the most critical issues per your business need. The status for a CVE is visible in all CVE tables in the vulnerability service and in individual CVE views.

3.9.1. Set a status for a CVE on all affected systems

You can set a status for a CVE and have that status apply to that CVE on all of the systems it impacts.

Procedure

1. Navigate to the Security > Vulnerability > CVEs tab and log in if necessary.
2. Click the **more-actions** icon located on the right end of the CVE row and select **Edit status**.
3. Select the appropriate status and, optionally, enter a reason for your decision in the **Justification** text box.
4. Select **Do not overwrite individual system status** if there are statuses set for this CVE on individual systems that you want to preserve.
   Otherwise, leave the box unchecked to apply this status to all of the systems it is impacting.
5. Click **Save**.

Additional resources

- Generating vulnerability service reports

Verification

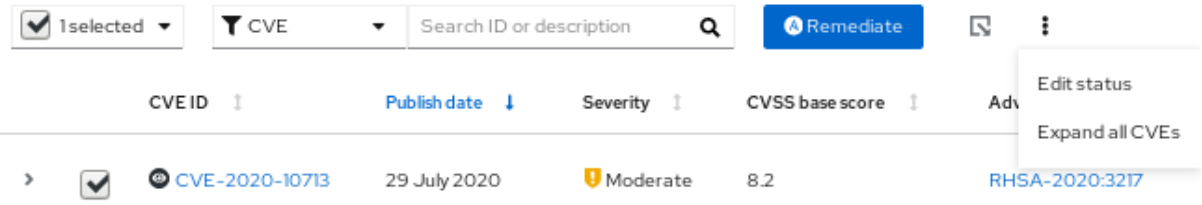
- Verify the CVE status is updated for all affected systems in the CVE details view.

3.9.2. Set a status for a CVE and system pair

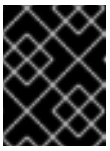
You can set a status on a CVE and system pair by using vulnerability service.

Procedure

1. Navigate to the Security > Vulnerability > Systems tab and log in if necessary.
2. Identify the system and click the system name to open it.
3. Select a CVE from the list and check the box next to the CVE ID.
4. Click the **more-options** icon in the toolbar and select **Edit status**.



5. In the pop-up card, take the following actions:
   a. Set a status for the CVE and system pair.



IMPORTANT

If you select the box **Use overall CVE status**, you cannot set a status for the pair.

   b. Optionally, enter a justification for your status determination.
   c. Click menu:[Save].

Verification

- Locate the CVE in the list and verify the status is set.

Additional resources

- Generating vulnerability service reports

3.10. USE THE SEARCH BOX

The search function in the vulnerability service works in the context of the page you are viewing.

CVEs page

The search box is in the toolbar at the top of the CVEs list. With the CVE filter set, search CVE IDs and descriptions.



Systems page

The search box is in the toolbar at the top of the list. Search for system name or UUID.



3.11. SORT CVE LIST DATA

You can use the sorting functions in the vulnerability service. The fields by which you can sort differ based on the context of the page you are viewing.

Procedure

1. In the **CVEs tab**, sort data by the following columns:
   - CVE ID
   - Publish date
   - Severity
   - CVSS base score
   - Systems exposed
   - Business risk
   - Status
2. In the **Systems tab**, sort data by the following columns:
   - Name
   - Applicable CVEs
   - Last seen
3. After selecting a system in the **Systems tab**, you can sort the system-specific list of CVEs by the following sorting options:
   - CVE ID
   - Publish date
   - Impact
   - CVSS base score
   - Business risk
   - Status

Additional resources

- Understanding severity ratings

- Generating vulnerability service reports

CHAPTER 4. ORGANIZE SYSTEMS IN RED HAT LIGHTSPEED BY USING FILTERING AND TAGS

Filter and organize your systems in Red Hat Lightspeed by using SAP workload tags, Satellite host group tags, and custom tags. Tagging enables you to focus on specific system groups across multiple services, which simplifies management for large-scale deployments.

4.1. FILTERS AND TAGS FOR SAP WORKLOADS

Red Hat Lightspeed automatically tags systems by SAP workload type and SAP ID (SID), which you can use to filter and organize SAP systems.

Use these automatic tags to filter and organize your SAP systems throughout the Red Hat Lightspeed application. No customization or manual tagging is required. To filter to view your SAP workloads, use **Workloads > SAP**.

4.2. FILTERS AND TAGS FOR SATELLITE HOST GROUPS

Satellite host groups are configured in Satellite and are automatically recognized by Red Hat Lightspeed.

4.3. CUSTOM SYSTEM TAGGING

Apply custom tags to your registered systems to add contextual markers and organize your Red Hat Lightspeed inventory. With custom tags, you can filter and focus on related groups of systems, which simplifies management for large-scale deployments.

You can also apply predefined tags to your systems. The advisor service uses predefined tags to generate targeted recommendations for systems that require specific attention, such as those with high security needs.

4.3.1. Filter structure

Filters use a **namespace=value** or **key=value** paired structure.

- **Namespace.** The namespace is the name of the ingestion point, *insights-client*. This value cannot be changed. The **tags.yaml** file is abstracted from the namespace, which is injected by the client before upload.
- **Key.** You can create the key or use a predefined key from the system. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.
- **Value.** You can define your own descriptive string value. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.

4.3.2. Create a custom group and the tags.yaml file

To create and add tags to **/etc/insights-client/tags.yaml**, use **insights-client** with the **--group=<name-you-choose>** option. Running this command on your systems that are registered to Red Hat Lightspeed completes the following actions:

- Creates the **etc/insights-client/tags.yaml** file

- Adds the **group=** key and **<name-you-choose>** value to **tags.yaml**
- Uploads a fresh archive from the system to the Red Hat Lightspeed service, making the new tag immediately visible along with your latest results

Prerequisites

- Root-level access to your system.

Procedure

1. Run the following command as root, adding your custom group name in place of **<name-you-choose>**:

```
[root@server ~]# insights-client --group=<name-you-choose>
```

2. Optional: To add additional tags, edit the **/etc/insights-client/tags.yaml** file.
3. Navigate to **Inventory → Systems** and log in if necessary.
4. Click the **Filter by tags** drop-down menu. You can also use the search box to enter all or part of the tag's name to automatically show systems with that text in the tags.
5. Scroll up or down the list to locate the tag.
6. Click the tag to filter by it.
7. Verify that your system is among the results on the advisor systems list.
   a. Navigate to **Inventory → Systems** and log in if necessary.
   b. Activate the **Name** filter and begin typing the system name until you see your system, then select it.
   c. The tag symbol is a darker color, and the number beside it shows the correct number of tags applied.

4.3.3. Add and modify custom system tags by editing tags.yaml

Edit the **tags.yaml** file to add or modify custom system tags after creating a **group** tag. Using custom system tags can help you streamline the organization and filtering of systems when you use Red Hat Lightspeed inventory and other services.

Prerequisites

- Root-level access to your system.

Procedure

1. Open the tag configuration file, **tags.yaml**, in an editor.

```
[root@server ~]# vim /etc/insights-client/tags.yaml
```

2. Edit the file contents or add **key=value** pairs as needed. Use a mix of capitalization, letters, numbers, symbols, and whitespace. Example **tags.yaml** with multiple tags:

```
group: _group-name-value_
location: _location-name-value_
description:
- RHEL8
- SAP
key 4: value
```

3. Save your changes and close the editor.

4. Initiate an upload to Red Hat Lightspeed.

```
[root@server ~]# insights-client
```

5. Navigate to **Inventory → Systems** and log in if necessary.

6. In the **Filter Results** box, click the down arrow and select one of the filters or enter the name of the filter and select it.



NOTE

You can search by the tag key or by its value.

7. Find your system among the results.

8. Verify that the filter icon is highlighted and shows a number representing the number of filters applied to the system.

4.4. PREDEFINED TAGS

In addition to custom tags, Red Hat Lightspeed provides predefined tags that the advisor service can use to create targeted recommendations. These predefined tags are useful for systems that require more security or need different networking performance levels.

To get the extended security hardening and enhanced detection and remediation capabilities offered by predefined tags, opt in and configure the tags. After configuration, the advisor service provides recommendations based on tailored severity levels and the preferred network performance that applies to your systems.

Use the **/etc/insights-client/tags.yaml** file to tag systems with predefined tags, similar to how you tag systems in the inventory service. The predefined tags are configured using the same **key=value** structure used to create custom tags. Details about the Red Hat-predefined tags are in the following table:

Table 4.1. List of supported predefined tags

Key	Value	Note

Key	Value	Note
security	**normal** (default) / **strict**	With the **normal** (default) value, the advisor service compares the system's risk profile to a baseline that comes from the default configuration of the most recent version of RHEL and from often-used usage patterns. This option keeps recommendations focused, actionable, and low in numbers. The **strict** value is best for systems that are security-sensitive. This option uses a stricter baseline and can show recommendations even on the most up-to-date RHEL installations.
network_performance	**null** (default) / **latency** / **throughput**	The network performance values, either latency or throughput according to your business requirement, affect the severity of an advisor service recommendation to a system.



NOTE

The names of predefined tag keys are reserved for specific use. If you already use the key **security**, with a value that differs from one of the predefined values, you do not see a change in your recommendations. You only see a change in recommendations if your existing **key=value** is the same as one of the predefined keys. For example, if you have a **key=value** of **security: high**, your recommendations do not change because of the predefined tags that Red Hat has already defined. If you currently have a **key=value** pair of **security: strict**, you see a change in the recommendations for your systems.

Additional resources

- Using system tags to enable extended security hardening recommendations
- Leverage tags to make Red Hat Lightspeed Advisor recommendations understand your environment better
- Custom system tagging

4.4.1. Configure predefined tags

Configure predefined tags to adjust recommendations and gain extended security hardening, enhanced detection, and remediation capabilities within the Red Hat Lightspeed advisor service.

Prerequisites

- You have root-level access to your system.
- You have insights-client installed.
- You have systems registered within the insights-client.
- You have created the **tags.yaml** file. For information about creating the **tags.yaml** file, see Configure custom group tags.

Procedure

1. Use the command line and your preferred editor to open **/etc/insights-client/tags.yaml**. (The following example uses Vim.)

```
[root@server ~]# vi /etc/insights-client/tags.yaml
```

2. Edit the **/etc/insights-client/tags.yaml** file to add the predefined **key=value** pair for the tags. This example shows how to add **security: strict** and **network_performance: latency** tags.

```
# cat /etc/insights-client/tags.yaml
group: redhat
location: Brisbane/Australia
description:
- RHEL8
- SAP
security: strict
network_performance: latency
```

3. Save your changes.
4. Close the editor.
5. Optional: Run the **insights-client** command to generate an upload to Red Hat Lightspeed, or wait until the next scheduled Red Hat Lightspeed upload.

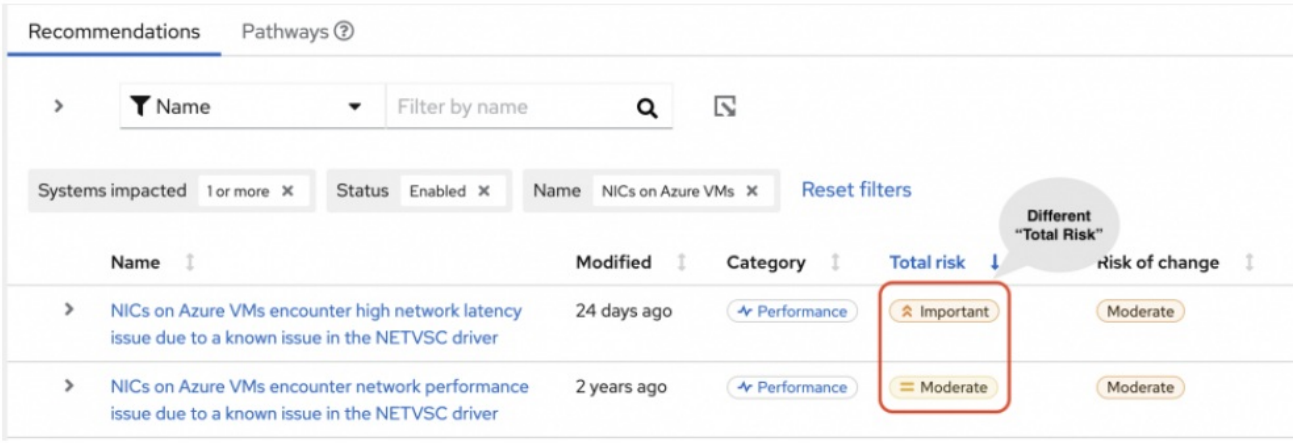
```
[root@server ~]# insights-client
```

Verification

After generating an upload to Red Hat Lightspeed (or waiting for the next scheduled Red Hat Lightspeed upload), you can discover whether the tags are in the production environment by accessing Red Hat Lightspeed > RHEL > Inventory > Systems . Find your system and look for the newly created tags. A table displays:

- Name
- Value
- Tag Source (for example, insights-client).

The following image of the advisor service shows a system with the **network_performance: latency** tag configured.



The system shows a recommendation with a higher Total Risk level of Important. The system without the **network_performance: latency** tag has a Total Risk of Moderate. You can make decisions about prioritizing the system with higher Total Risk.

4.5. ADDITIONAL RESOURCES

- Refining your view of systems in inventory
- Enable SQL Server recommendations in Advisor
- Group and tag filters in the compliance service

CHAPTER 5. REFERENCE MATERIALS

For more information about the vulnerability service, see the following resources:

- Generating vulnerability service reports with FedRAMP
- Red Hat Lightspeed Documentation
- Red Hat Lightspeed Product Support page